Computer and Network Security (2026-2027)

Part 4: Homework – Number Theory

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Divisibility

- We say that a nonzero b **divides** a (also written as b | a, or b is a **divisor** of a)
  - if $a = m \times b$ for some m
  - where a, b, and m are integers
- Properties of divisibility
  - If a | 1, then a = ±1
  - If a | b and b | a, then a = ±b
  - Any b ≠ 0 divides 0
  - If a | b and b | c, then a | c
  - If b | g and b | h, then b | (m g + n h) for arbitrary integers m and n

Greatest common divisor (GCD)

- The **greatest common divisor** of a and b is the largest integer that divides both a and b
  - Written as gcd(a, b)
- The greatest common divisor should be positive
  - gcd(a,b) = gcd(a,-b) = gcd(-a,b) = gcd(-a,-b)
  - Or, gcd(a,b) = gcd(|a|, |b|)
- Because all nonzero integers divide 0, gcd(a, 0) = |a|
- Two integers a and b are **relatively prime** if gcd(a, b) = 1

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Modulus definition

- If a is an integer and n is a positive integer, we define **a mod n** to be the remainder when a is divided by n; the integer n is called the **modulus**
- Thus, for any integer a:
  - $a = qn + r \quad 0 \le r < n;\ q = [a/n]$
  - $a = [a/n] * n + (a \bmod n)$
- Two integers a and b are said to be **congruent modulo n** if (a mod n) = (b mod n)
  - This is written as $a \equiv b \pmod n$
  - Note that if $a \equiv 0 \pmod n$, then $n \mid a$
- **Congruences** have the following properties:
  - $a \equiv b \pmod n$ if $n = (a - b)$
  - $a \equiv b \pmod n$ implies $b \equiv a \pmod n$
  - $a \equiv b \pmod n$ and $b \equiv c \pmod n$ imply $a \equiv c \pmod n$

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Additional properties of modular arithmetic

- [(a mod n) + (b mod n)] mod n = (a + b) mod n
- [(a mod n) - (b mod n)] mod n = (a - b) mod n
- [(a mod n) * (b mod n)] mod n = (a * b) mod n
- Inverse
  - a is the additive inverse of b if $(a + b) \bmod n = 0$
  - a is the multiplicative inverse of b if $a \times b \equiv 1 \pmod{n}$

Property	Expression
Commutative Laws	$(w + x) \bmod n = (x + w) \bmod n$ $(w \times x) \bmod n = (x \times w) \bmod n$
Associative Laws	$[(w + x) + y] \bmod n = [w + (x + y)] \bmod n$ $[(w \times x) \times y] \bmod n = [w \times (x \times y)] \bmod n$
Distributive Law	$[w \times (x + y)] \bmod n = [(w \times x) + (w \times y)] \bmod n$
Identities	$(0 + w) \bmod n = w \bmod n$ $(1 \times w) \bmod n = w \bmod n$
Additive Inverse ($-w$)	For each $w \in Z_n$, there exists a z such that $w + z \equiv 0 \bmod n$

Prime number properties

- Prime numbers only have as divisors 1 and themselves
- They cannot be written as a product of other numbers
- Prime numbers are central to number theory

- Any integer a > 1 can be factored in a unique way as
  - $a = p_1^{a_1} \times p_2^{a_2} \times \cdots \times p_t^{a_t}$
  - where $p_1 < p_2 < \ldots < p_t$ are prime numbers and where each a_i is a positive integer
  - This is known as the **fundamental theorem of arithmetic**

- Given $a = \prod_{p \in P} p^{a_p}$ and $b = \prod_{p \in P} p^{b_p}$ then
  - $k = a \times b = \prod_{p \in P} p^{a_p + b_p}$ (with P the set of all prime numbers)
  - if a | b then $a_p \leq b_p$ for all p.

Multiplicative inverse

- The multiplicative inverse of a is a^{-1} if: $a \times a^{-1} \equiv 1 \pmod{m}$
- The value of a^{-1} can be found by using the **Extended Euclidian Algorithm**:

Initialization

$r_{-1} = m$ $\quad$ $w_{-1} = 0$

$r_0 = a$ $\quad$ $w_0 = 1$

Recursive algorithm

$r_i = r_{i-2} \bmod r_{i-1}$

$q_i = r_{i-2} / r_{i-1}$

$w_i = w_{i-2} - q_i \, w_{i-1}$

Stop condition

if $r_i = 1$ then $w_i = a^{-1}$

Fermat's theorem

- Fermat's and Euler's theorems are very important in public key cryptography
- If p is prime and a is a positive integer not divisible by p, then
$$a^{p-1} \equiv 1 \pmod{p}$$
- (alternative form) If p is a prime and a is a positive integer then
$$a^p \equiv a \pmod{p}$$

Euler's theorem

- **Euler's Totient Function $\phi(n)$** is defined as the number of positive integers less than n that are relatively prime to n
- For every a and n that are relatively prime

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

- Alternative form

$$a^{\phi(n)+1} \equiv a \pmod{n}$$

- Examples of Euler's Totient Function:

n	$\phi(n)$
1	1
2	1
3	2
4	2
5	4
6	2
7	6
8	4
9	6
10	4

n	$\phi(n)$
11	10
12	4
13	12
14	6
15	8
16	8
17	16
18	6
19	18
20	8

n	$\phi(n)$
21	12
22	10
23	22
24	8
25	20
26	12
27	18
28	12
29	28
30	8

Miller-Rabin Algorithm

Used to test large integers for primality (with a certain probability)

Prime(n)?

1. Find integers k, q, with k > 0, q odd, so that (n – 1) = 2^k q ;
2. Select a random integer a, 1 < a < n – 1 ;
3. if a^q mod n = 1 then return ("n potentially prime") ;
4. for j = 0 to k – 1 do
5. if ($a^{2^j q} \ mod \ n = n - 1$) then return ("n potentially prime") ;
6. return ("n not prime") ;

Repeat t times to reduce false positives to $(1/4)^t$

False positive rate: 1/4

Chinese remainder theorem (CRT)

Let

$$M = \prod_{i=1}^{k} m_i$$

where the m_i are pairwise relatively prime

We can represent any integer $A \in \mathbb{Z}_M$ by a k-tuple whose elements are in $\mathbb{Z}_{m_i}$ using the correspondence

$$A \leftrightarrow (a_1, a_2, \ldots, a_k)$$

where $a_i \in \mathbb{Z}_{m_i}$ and $a_i = A \bmod m_i$

$\mathbb{Z}_x$ is the set of integers modulo x

Calculating A based on its unique bijection

For every A there is a unique mapping to a k-tuple, called a bijection

We can calculate A from the k-tuple (a_1, …, a_k) as follows:

$$A \equiv \left(\sum_{i=1}^{k} a_i c_i \right) (mod\ M)$$

with

$$c_i = M_i \times \left(M_i^{-1} mod\ m_i \right)$$

and

$$M_i = {}^{M}/_{m_i}$$

Operations on A can be performed on bijection

Operations performed on A can be equivalently performed individually on each element of the corresponding k-tuple

Given:

$$A \leftrightarrow (a_1, a_2, \dots, a_k) \text{ and } B \leftrightarrow (b_1, b_2, \dots, b_k)$$

then

$$(A + B)\ mod\ M \ \leftrightarrow \left((a_1 + b_1)\ mod\ m_1, \cdots, (a_k + b_k)\ mod\ m_k\right)$$
$$(A - B)\ mod\ M \ \leftrightarrow \left((a_1 - b_1)\ mod\ m_1, \cdots, (a_k - b_k)\ mod\ m_k\right)$$
$$(A \times B)\ mod\ M \ \leftrightarrow \left((a_1 \times b_1)\ mod\ m_1, \cdots, (a_k \times b_k)\ mod\ m_k\right)$$

As such, the CRT provides a way to **manipulate large numbers mod M** in terms of tuples of smaller numbers.

Primitive root

Consider the general expression: $a^m \equiv 1 \pmod{n}$

- According to Euler's theorem, if a and n are relatively prime, then there is at least 1 integer m that satisfies this, namely $m = \phi(n)$
- The lowest positive exponent m for which this holds is to **order** of a (mod n)
- A number *a* is a **primitive root** of *n* if the order of a (mod n) is $\phi(n)$
  - Then $a, a^2, \dots, a^{\phi(n)}$ are distinct (mod n) and all relatively prime to n
  - For a prime number p, if a is a primitive root of p, then $a, a^2, \dots, a^{p-1}$ are all distinct (mod p).
  - Only $2, 4, p^\alpha, 2p^\alpha$ (with p a prime, and α a positive integer) have primitive roots

Discrete logarithms for modular arithmetic

- For a primitive root *a* of a prime *p*, the powers of a from 1 to (p - 1) produce each integer from 1 through (p – 1) exactly once
- We also know that for any integer *b* an *r* exists, such that: $b \equiv r \pmod{p}$, with $0 \leq r \leq p-1$

The two points above, prove the existence of a unique exponent i:

$$b \equiv a^i \pmod{p} \quad \text{where } 0 \leq i \leq (p-1)$$

The exponent i is referred to as the **discrete logarithm** of the number *b* for the base *a (mod p)*, denoted as $\text{dlog}_{a,p}(b) = i$

- $\text{dlog}_{a,p}(1) = 0$
- $\text{dlog}_{a,p}(a) = 1$

- Used in **Diffie-Hellman** and **Elgamal** encryption

Example of discrete logarithms (1)

Example: p = 7

Step 1: Find primitive root(s) of 7

$a \in [1,6]$ is a primitive root of 7 if the smallest m for which $a^m \equiv 1 \pmod{7}$ equals $\phi(7) = 6$:

- a = 1: smallest m = 1
- a = 2: smallest m = 3
- a = 3: smallest m = 6
- a = 4: smallest m = 3
- a = 5: smallest m = 6
- a = 6: smallest m = 2

Both 3 and 5 are primitive roots of 7

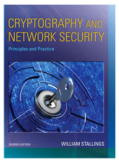
Example of discrete logarithms (2)

Example: p = 7

Step 2: Find discrete logarithms for a = 3 and p = 7

- $3^1 \ mod \ 7 = 3 \quad \Rightarrow dlog_{3,7}(3) = 1$
- $3^2 \ mod \ 7 = 2 \quad \Rightarrow dlog_{3,7}(2) = 2$
- $3^3 \ mod \ 7 = 6 \quad \Rightarrow dlog_{3,7}(6) = 3$
- $3^4 \ mod \ 7 = 4 \quad \Rightarrow dlog_{3,7}(4) = 4$
- $3^5 \ mod \ 7 = 5 \quad \Rightarrow dlog_{3,7}(5) = 5$
- $3^6 \ mod \ 7 = 1 \quad \Rightarrow dlog_{3,7}(1) = 6$

Further reading



Link with the book

- Chapter 2 (Sections 2.1 – 2.8)